

AWS Cloud Environment Design for Development Teams

Project Name: Cloud Enablement for Engineering Teams

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Objective

To design and implement a secure, scalable, and maintainable AWS cloud environment for the development and engineering teams at Data Lake Limited, integrating with the corporate network and existing Microsoft Active Directory.

Architecture Overview

Key Components:

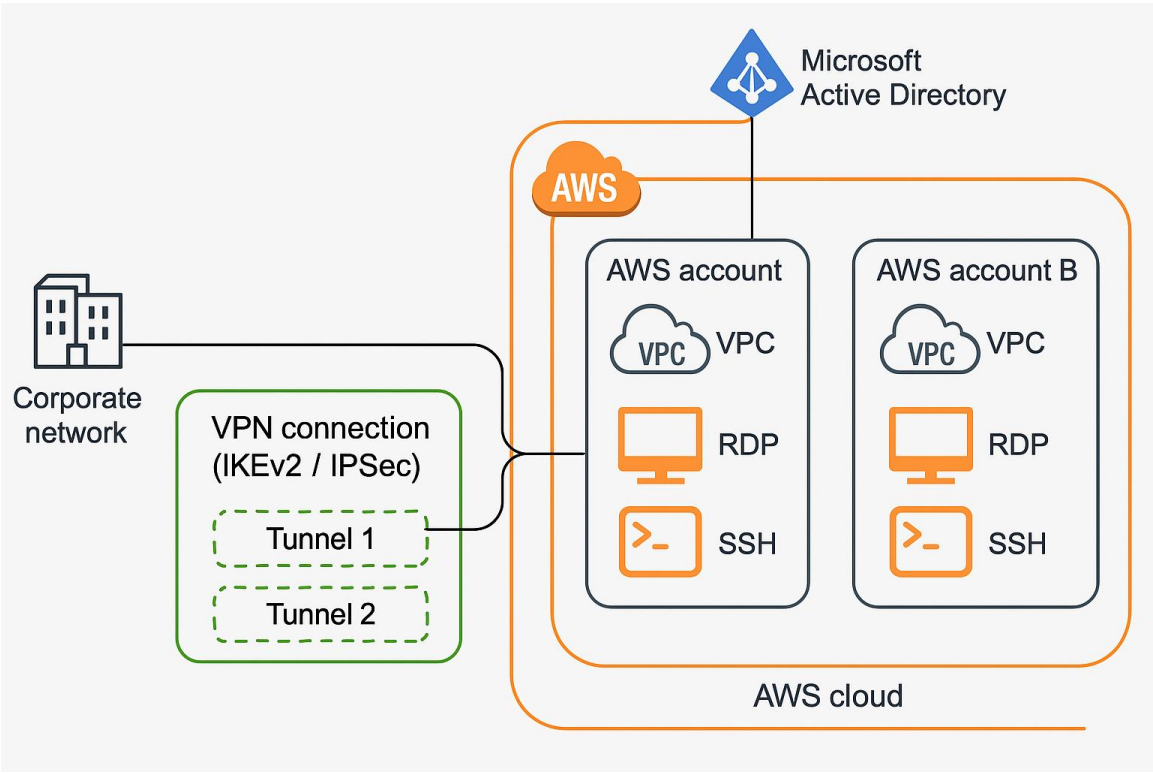
- - AWS Organizations with multiple accounts: Management, Shared Services, Dev Team 1, Dev Team 2
- - VPCs in each account with public and private subnets
- - VPN Connectivity between on-prem Data Center (Preston) and AWS using Site-to-Site VPN with IKEv2/IPSec and dual redundant tunnels
- - AWS Directory Service (Enterprise or AD Connector) integrated with on-prem AD (duu.local and blue.net)
- - SSO Integration via IAM Identity Center
- - Secure developer access (RDP/SSH) to EC2 resources
- - Region: AWS London (eu-west-2)

Security Considerations

- - End-to-end IPSec encryption on VPN tunnels
- - SSO using existing Active Directory credentials
- - MFA handled on endpoint devices (no additional MFA in AWS)
- - Use of IAM roles tied to AD groups
- - EC2 instances protected via Security Groups and NACLs
- - Optional: Use AWS Systems Manager for secure, portless session access

Networking Diagram

[Insert link or image of the diagram uploaded to Draw.io or Confluence]



Implementation Plan

Step	Task	Owner	Status
1	Create AWS Organization and accounts	Cloud Team	☐
2	Setup VPCs in each account	Cloud Team	☐
3	Deploy VPN Gateway, configure Site-to-Site VPN	Network Team	☐
4	Deploy AWS Directory Service or AD Connector	Cloud Team	☐
5	Integrate SSO with Microsoft AD	IAM Admin	☐

6	Configure EC2 access (RDP/SSH)	Cloud Team	☐
7	Review with Enterprise Architecture Team	Cloud Architect	☐
8	Submit for InfoSec Approval	Cloud Architect	☐
9	Begin environment build with Cloud Engineers	Cloud Team	☐

Review & Approvals

- - Enterprise Architecture Team – Alignment on account strategy and connectivity
- - Information Security Team (InfoSec) – Approval of VPN, AD, and access control model